

Revised Solution for P2-(b)

(1) $\vec{E} = (-\frac{16}{r^2}) \hat{a}_r [\frac{V}{m}]$, $\vec{E}$ is a function of r .

$\vec{D} = \epsilon_0 \vec{E} = -\epsilon_0 \frac{16}{r^2} \hat{a}_r [\frac{C}{m^2}]$. $\vec{D}$ at the point B

$A = (2\pi, \pi, \frac{\pi}{2}) \rightarrow B = (4m, 0, \pi)$, where, $r = 4[m]$

Therefore,

$$\vec{D}_B = -\left(\frac{1}{36\pi} \times 10^{-9}\right) \frac{16}{(4)^2} \hat{a}_r = -\frac{10^{-9}}{36\pi} \hat{a}_r [\frac{C}{m^2}]$$
$$\underline{\underline{D_B = -\frac{10^{-9}}{36\pi}}}$$

(2) $V_{AB} = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{r} = \int_B^A \vec{E} \cdot d\vec{r}$

$$= \int_4^2 \left(-\frac{16}{r^2}\right) dr = \underline{\underline{-4V}}$$